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Medical Appointment No-Shows

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I. Introduction.....	1
1.1. Problem Statement.....	1
1.2. Objective.....	1
II. Dataset.....	2
III. Data Preprocessing.....	3
3.1. Data Cleaning Steps.....	3
3.2. Data Quality Issues Addressed.....	3
3.3. Final Dataset Structure.....	4
IV. EDA.....	5
4.1. Exploratory Data Analysis.....	5
4.2. Univariate Analysis.....	5
4.3. Bivariate Analysis.....	9
4.4. Statistical Test.....	12
V. Methodology.....	12
5.1. Advanced Analysis.....	12
5.1.1 Principal Component Analysis (PCA).....	12
5.2. Clustering Analysis.....	13
5.2.1 K-means Clustering.....	13
5.2.2 Hierarchical Clustering.....	14
5.3. Gaussian Mixture Models (GMM).....	15
5.3.1 Model Specification.....	15
5.3.2 Component Interpretation.....	18
5.4. Correspondence Analysis (CA).....	20
5.4.1. Methodology Results.....	20
VI. Conclusion.....	22
References.....	23

I. Introduction

Missed medical appointments, or no-shows, pose a significant challenge within Brazil's public healthcare system (SUS), leading to wasted resources, longer wait times, and reduced operational efficiency. Understanding the underlying factors is essential for improving service delivery. This study employs advanced analytical methods—including Principal Component Analysis (PCA), K-means and Hierarchical Clustering, Gaussian Mixture Models (GMM), and Correspondence Analysis (CA)—to uncover patterns in patient behavior and identify key drivers of appointment absenteeism. The aim is to provide actionable insights to help healthcare providers optimize scheduling and reduce no-show rates.

Team contributions include: SORN Sreynatt (Data Preprocessing and Cleaning), ROEUN Sovandeth (Exploratory Data Analysis and Visualization), and SAN Kimheang (Modeling and Advanced Analysis including PCA, Clustering, and Predictive Models).

1.1. Problem Statement

Appointment no-shows in Brazil's public healthcare system result in five key problems:

- 1. **Resource Waste** – Unused clinical time, staff effort, and facilities.
- 2. **Financial Losses** – Costs incurred without delivered care.
- 3. **Access Barriers** – Longer wait times for other patients.
- 4. **Health Impacts** – Potential worsening of patient conditions.
- 5. **Operational Inefficiency** – Complicated scheduling and reduced productivity

1.2. Objective

The main objectives of this study are to understand and improve appointment attendance in Brazilian public healthcare.

1. **Identify factors** Analyze patient and appointment characteristics to understand why some patients miss their appointments.
2. **Segment high-risk patients** Group patients based on no-show risk to target interventions effectively.
3. **Provide actionable recommendations** Suggest strategies like reminders or optimized scheduling to reduce no-shows and improve clinic efficiency.

4. **Apply all statistical techniques** including PCA, clustering, GMM, and CA. Use PCA, clustering, GMM, and correspondence analysis to uncover patterns and relationships in patient behavior.

II. Dataset

Dataset Overview: 110,527 appointment records from Brazilian public healthcare centers (May 2016), containing patient demographics, health conditions, and appointment details.

FIRST 5 ROWS OF MEDICAL APPOINTMENT DATASET

	PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap	SMS_received	No-show
0	2.987250e+13	5642983	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	0	0	No
1	5.589978e+14	5642583	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	0	0	No
2	4.262962e+12	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	62	NATA DA PRAIA	0	0	0	0	0	0	No
3	8.679512e+11	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0	0	0	0	0	0	No
4	8.841186e+12	5642494	F	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	0	0	No

Variable Dictionary

- **PatientId** - Unique patient identifier
- **AppointmentID** - Unique appointment identifier
- **Gender** - Patient gender (F/M)
- **ScheduledDay** - Appointment scheduling timestamp
- **AppointmentDay** - Appointment date (time always 00:00:00)
- **Age** - Patient age in years
- **Neighbourhood** - Healthcare facility location
- **Scholarship** - Welfare program enrollment
- **Hipertension** - Hypertension diagnosis
- **Diabetes** - Diabetes diagnosis
- **Alcoholism** - Alcohol use disorder
- **Handcap** - Disability level (0-4)
- **SMS_received** - Appointment reminder sent
- **No-show** - Appointment outcome

1. Initial Observations

- **Complete data** - No missing values were detected, ensuring data integrity.
- **Time limitation** - Only appointment dates are available; exact times are missing.
- **Binary encoding** - Health conditions are correctly coded as 0 (absent) or 1 (present).
- **Geographic diversity** - Patients come from 81 different neighborhoods.

- **Temporal anomalies** - A few scheduling dates occur after appointment dates, indicating data inconsistencies.

2. Summary Statistics

The dataset provides a detailed overview of patient appointments and their characteristics. Key statistics are as follows:

- Total records: 110,527 appointments
- Variables: 14 original features
- Age range: 0-115 years
- Neighbourhoods: 81 unique locations
- Health condition prevalence:
 - a. Hypertension: 19.7% (21,801 patients)
 - b. Diabetes: 7.2% (7,943 patients)
 - c. Alcoholism: 3.0% (3,360 patients)
 - d. Any disability: 2.0% (2,241 patients)

III. Data Preprocessing

3.1. Data Cleaning Steps

The dataset was cleaned and preprocessed to ensure analytical integrity:

- **Completeness Check:** No missing values found.
- **Duplicate Removal:** All appointment records were unique.
- **Data Validation:** One invalid record (age = -1) was removed.
- **Date Standardization:** Scheduled and appointment dates were converted to datetime format for temporal analysis.
- **Consistency Checks:** Variables were verified for logical consistency, and binary codes were confirmed.

3.2. Data Quality Issues Addressed

Quality Dimension	Assessment	Action Taken
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Completeness	Excellent	No missing values found
Accuracy	Good	Fixed temporal inconsistencies
Validity	Good	Removed invalid age record
Consistency	Good	Standardized date formats
Uniqueness	Excellent	No duplicate rows

The dataset was evaluated for quality and found to be highly reliable. It showed excellent completeness with no missing values and strong uniqueness with no duplicate records. Minor issues, such as scheduling-after-appointment inconsistencies and one invalid negative age, were corrected. Date formats were standardized for consistency. These steps prepared the dataset for robust and accurate analysis.

3.3. Final Dataset Structure

Post-Cleaning Dataset Characteristics:

- **Total Records:** 110,526 valid medical appointments
- **Original Variables:** 14 preserved features
- **Engineered Features:** 9 new variables created
- **Final Dataset:** 23 total variables

Engineered Features Created:

1. **AwaitingDays:** Calculated days between scheduling and appointment
2. **AgeGroup:** Five categorical age groups (Child, Teen, Young Adult, Adult, Senior)
3. **IsChild:** Binary flag for patients under 18 years
4. **IsSenior:** Binary flag for patients 60+ years
5. **TotalConditions:** Sum of all health conditions per patient
6. **HasChronicCondition:** Flag for any chronic health condition
7. **HasMultipleConditions:** Flag for multiple comorbidities
8. **NoShow_binary:** Binary encoded target variable (0=Show, 1=No-show)
9. **WaitingCategory:** Five-category waiting period classification

Key Insights from Data Processing:

- **Dataset Imbalance:** 20.2% no-show rate (22,319 appointments) vs. 79.8% show rate
- **Waiting Time Impact:** Significant correlation found between waiting time and no-show probability
- **Health Burden:** 23.9% of patients have chronic conditions, 7.5% have multiple conditions
- **Appointment Patterns:** 39.6% of appointments scheduled for same day

IV. EDA

4.1. Exploratory Data Analysis

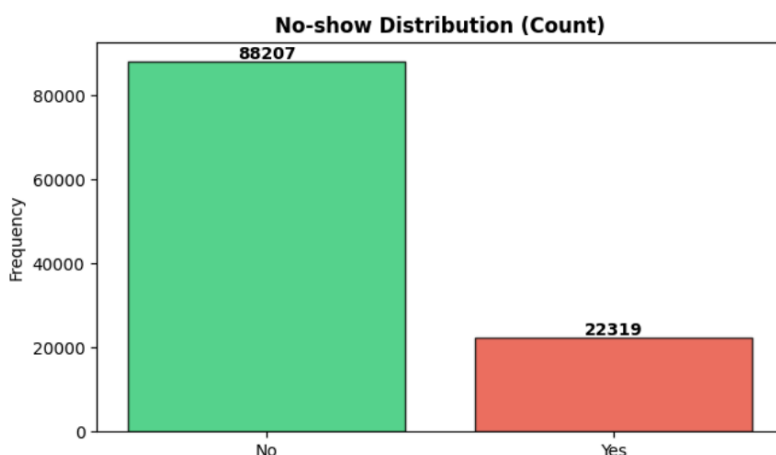
The exploratory data analysis aimed to understand the factors influencing patient no-shows and to identify patterns in appointment scheduling. Analysis of the target variable revealed an imbalance, with most patients attending their appointments, highlighting the need to carefully evaluate metrics beyond accuracy, such as recall and AUC. Demographic features, including age, gender, and scholarship status, were explored to examine their association with no-show behavior. EDA helps in:

- Identifying important features for modeling (like age, waiting days, SMS reminders).
- Detecting imbalances or outliers in the target variable (No-show).
- Understanding categorical patterns (gender, neighbourhood, day of week).
- Visualizing appointment timing trends (weekdays, time of day).

4.2. Univariate Analysis

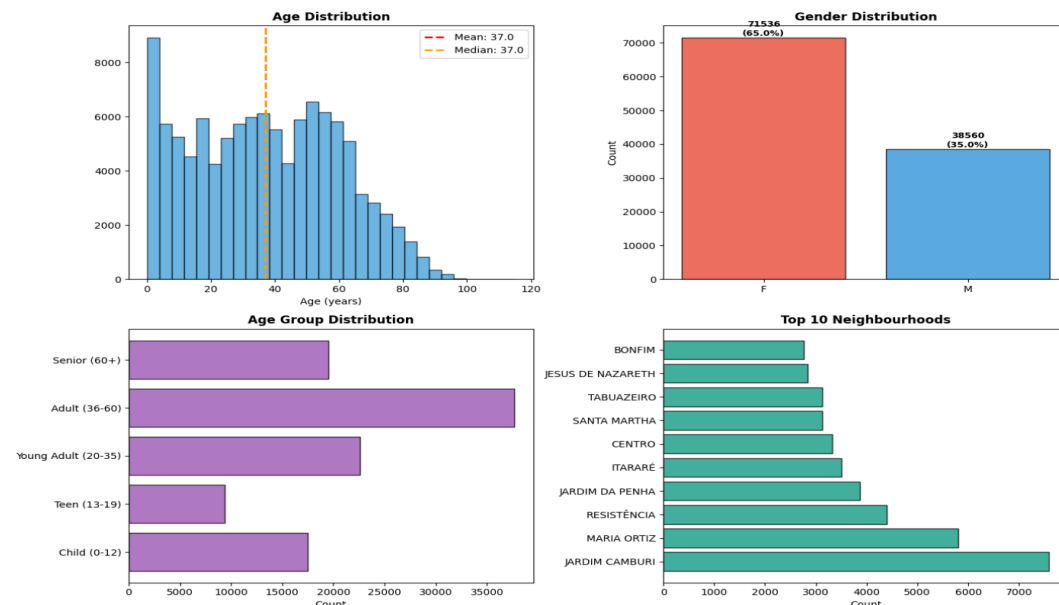
Univariate analysis was conducted to examine the distribution and characteristics of individual variables in the dataset. To better capture temporal patterns associated with patient attendance, several datetime-based features were created from the appointment data. The number of appointments in each time period was counted to see when most patients scheduled their visits. These features help the model understand time-related patterns in no-shows.

Target Variable: No-show Distribution



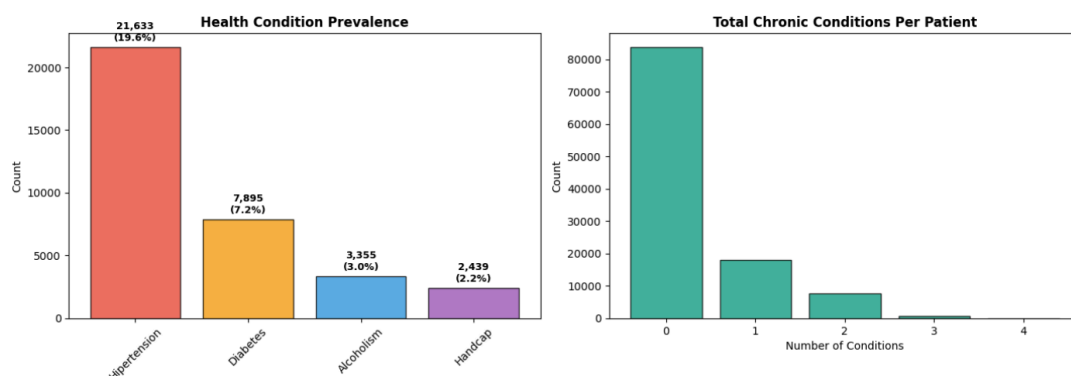
The distribution of the target variable, No-show, was analyzed to understand how many patients attended their appointments versus those who missed them. Out of the total appointments, 88,207 patients (approximately 79.8%) attended their appointments, while 22,319 patients (approximately 20.2%) did not show up. This indicates a class imbalance, with almost four times more patients attending than missing appointments, which should be considered during predictive modeling.

Demographic Variables



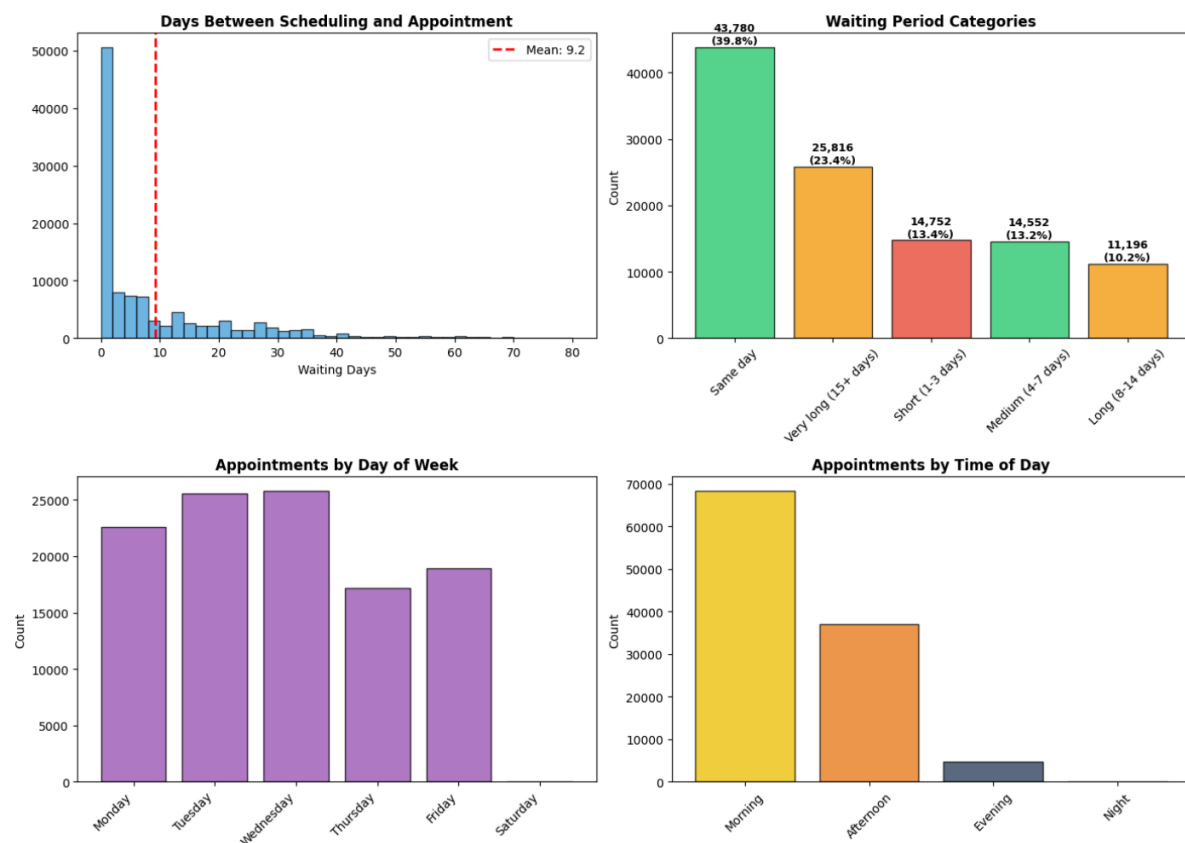
The average patient age is 37 years, equal to the median, indicating a balanced age distribution, with a standard deviation of 23 years reflecting wide age diversity. Females account for 65% of appointments, while males make up 35%, showing higher female attendance. By age group, adults (36–60) form the largest segment, followed by young adults (20–35), with seniors (60+) and children (0–12) also represented, and teenagers (13–19) being the smallest group. Overall, the patient population is diverse but primarily concentrated among adult age groups.

Health Condition Prevalence



Most patients in the dataset are healthy. 83,858 patients (76.17%) have no chronic conditions, while 26,238 patients (23.83%) have at least one chronic illness. Hypertension is the most common condition, affecting 21,633 patients (19.65%). Diabetes affects 7,895 patients (7.17%), followed by alcoholism with 3,355 patients (3.05%). Physical disability is the least common, seen in 2,439 patients (2.22%). Regarding comorbidity, 18,011 patients (16.36%) have one chronic condition, and 8,227 patients (7.47%) have more than one. These patients may find it harder to attend appointments. This shows that health conditions are an important factor when studying no-show behavior

Appointment Characteristics

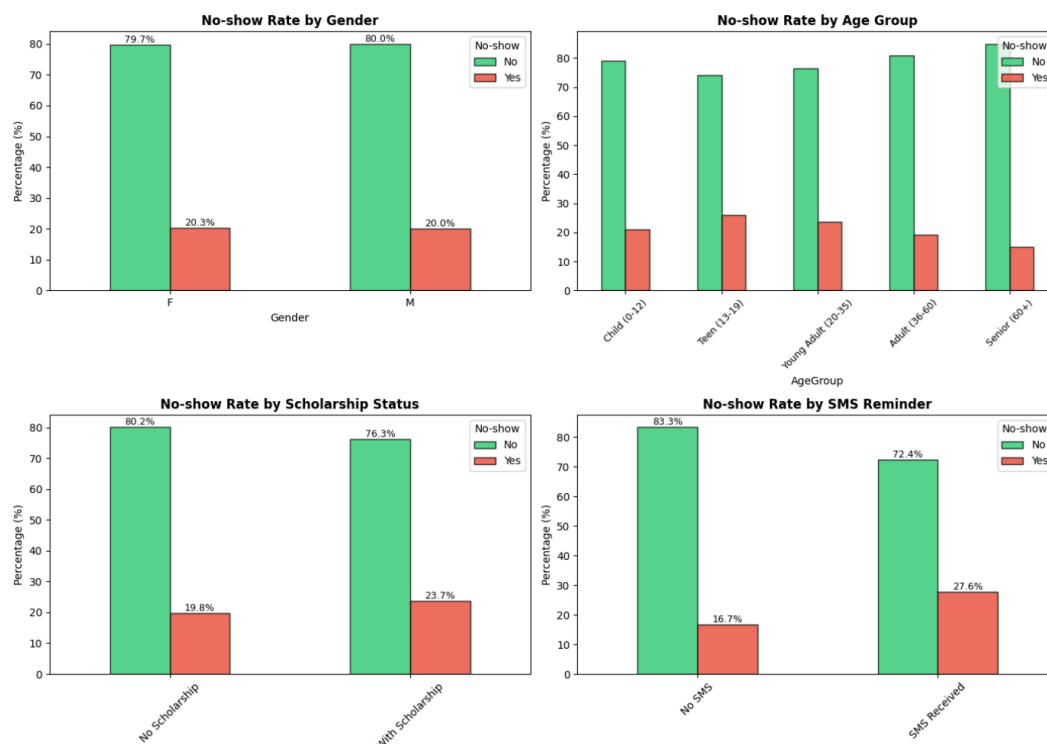


Most appointments have a short waiting time. The average waiting time is 9 days, but half of the patients wait 3 days or less. Some patients wait much longer, up to 80 days. Same-day appointments are the most common (39.77%). However, many patients (23.45%) wait 15 days or more, which may increase the chance of no-shows. Almost all appointments happen on weekdays, with very few on weekends. Only 31.96% of patients received an SMS reminder, which could affect whether patients attend their appointments

4.3. Bivariate Analysis

Bivariate analysis examines the relationship between two variables to see how one may affect the other. In this study, we explored how patient characteristics and appointment details relate to no-show behavior. For example, we looked at how age, gender, health conditions, waiting time, day of the week, and SMS reminders are associated with whether a patient attended their appointment.

No-show vs Demographic Factors:

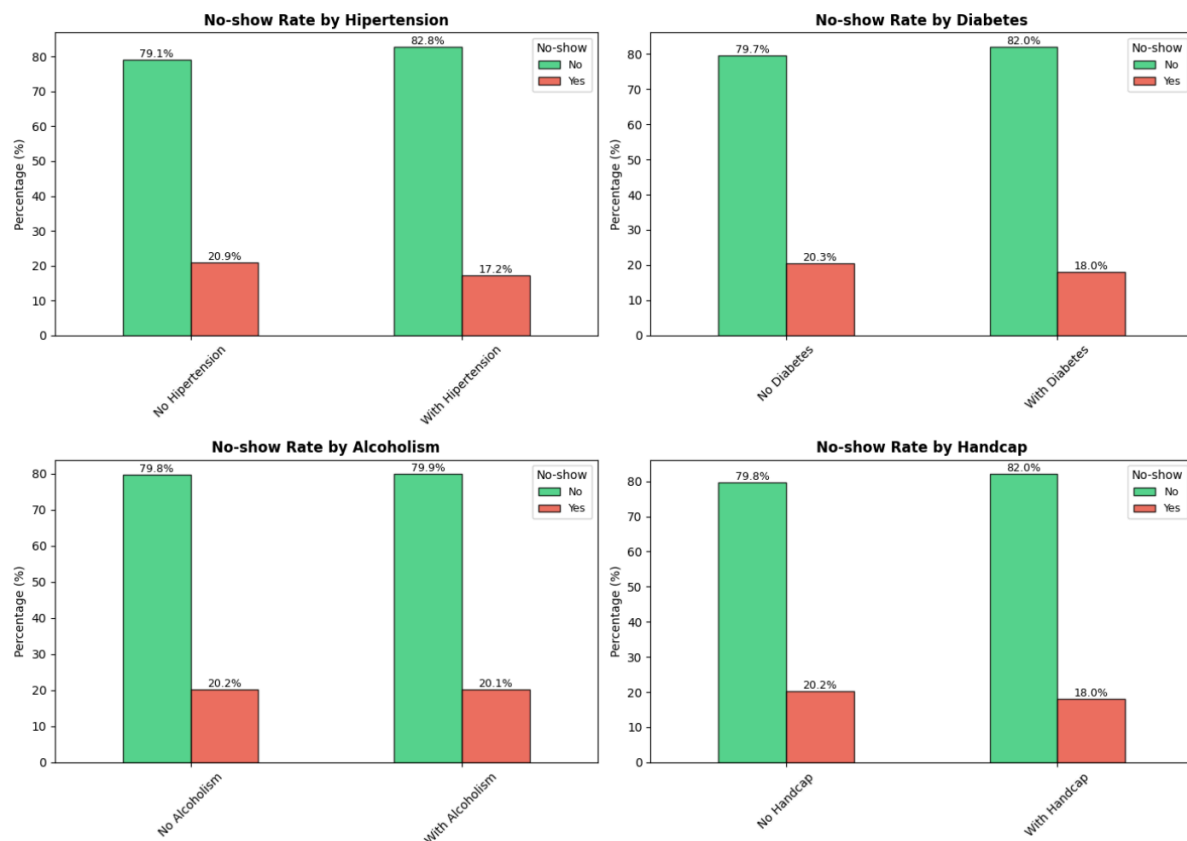


The analysis shows how no-show rates vary by gender, age group, scholarship status, and SMS reminders.

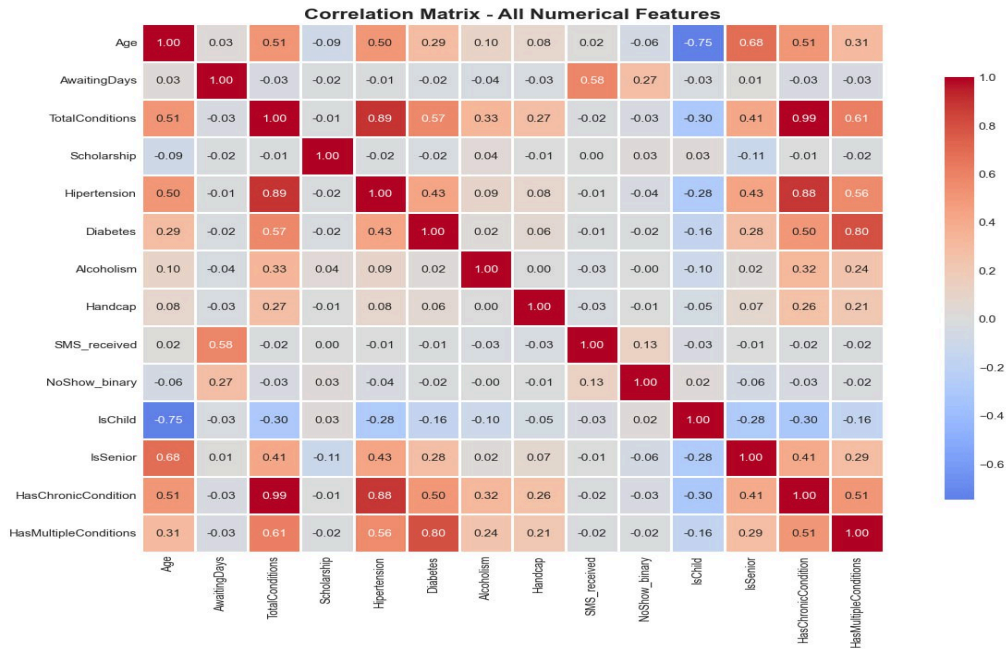
- Gender: Female patients have a slightly higher no-show rate (20.29%) compared to male patients (19.95%). This indicates that gender differences exist, but they are small.
- Age Group: No-show rates vary more clearly across age groups. Teenagers (13–19 years) have the highest no-show rate (25.95%), followed by young adults (20–35 years) at 23.7%. Children (0–12 years) show 20.93%, adults (36–60 years) 19.08%, and seniors (60+ years) have the lowest no-show rate (15.13%). This suggests that younger patients are more likely to miss appointments, while older patients are more reliable.

These results indicate that age is an important factor in predicting no-show behavior, and gender has a smaller effect.

No-show vs Health Factors



The analysis shows how chronic health conditions relate to no-show rates. Patients with hypertension have a no-show rate of 17.21%, while patients with diabetes have a slightly higher rate at 17.96%. Patients with alcoholism show the highest no-show rate among the conditions at 20.12%, and those with physical disability (handicap) have a no-show rate of 17.74%. Overall, patients with health conditions tend to have slightly lower or similar no-show rates compared to the general patient population, suggesting that having a chronic condition does not strongly increase the likelihood of missing appointments. However, alcoholism appears to have a small but noticeable effect on attendance. This information is important for understanding how patient health factors influence appointment adherence.



Correlation analysis shows that AwaitingDays has a small positive correlation with no-show (0.27), meaning longer waits slightly increase missed appointments. Age, TotalConditions, HasChronicCondition, and HasMultipleConditions have very weak correlations with no-show. IsChild and IsSenior are strongly correlated with age and weakly negatively correlated with no-show, indicating children and seniors are slightly less likely to miss appointments. Health factors are highly correlated with each other but show minimal effect on no-show. SMS_received has a small positive correlation (0.13), possibly because reminders are sent to high-risk patients. Overall, no single feature strongly predicts missed appointments.

4.4. Statistical Test

To examine the relationship between categorical variables and No-show, Chi-square tests of independence were performed. The tested variables were Gender, Scholarship, SMS_received, WaitingCategory, and AgeGroup. The results show:

- Gender: Not significant ($p = 0.18$), indicating no strong association with no-shows.
- Scholarship: Significant ($p \approx 2.68e-22$), suggesting that patients with scholarships are more likely to have different attendance patterns.
- SMS_received: Highly significant ($p \approx 0$), indicating that receiving an SMS reminder is strongly associated with attendance.
- WaitingCategory: Highly significant ($p \approx 0$), meaning the length of waiting time is strongly linked to whether a patient shows up.

- AgeGroup: Highly significant ($p \approx 4.27e-152$), showing that age categories influence appointment attendance.

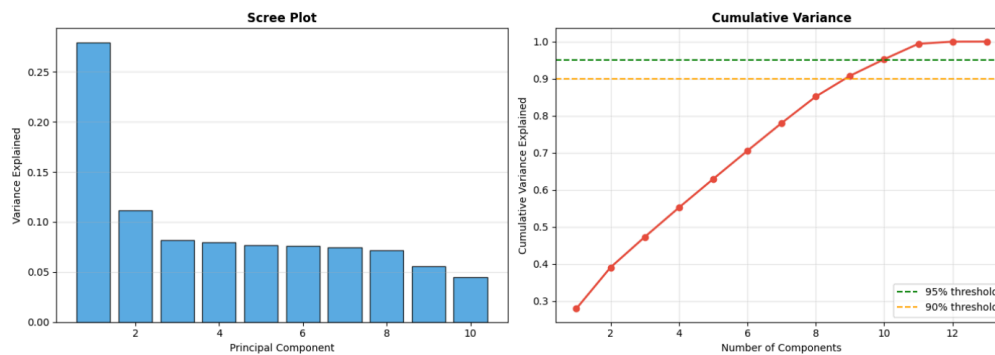
In summary, most categorical variables, except Gender, show a significant association with No-show, meaning they can help explain differences in patient attendance. Variables like SMS reminders, waiting time, and age group are particularly important in understanding no-show behavior.

V. Methodology

5.1. Advanced Analysis

5.1.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to the numerical variables to reduce dimensionality and identify underlying patterns in the data. Before applying PCA, all numeric features were standardized using z-score normalization to ensure equal contribution from each variable. PCA transforms the original correlated variables into a smaller set of uncorrelated principal components, ordered by the amount of variance they explain. The explained variance ratio and cumulative variance were used to determine how many components are needed to retain most of the information.

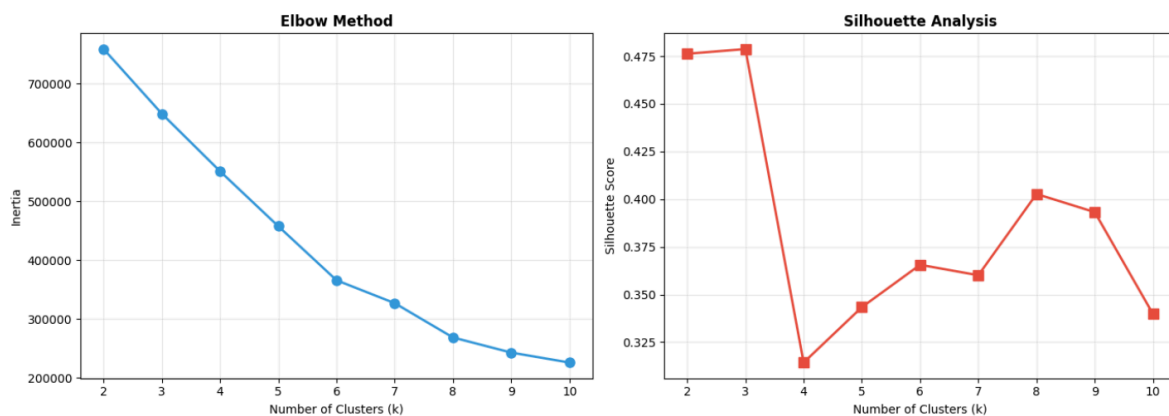


Principal Component Analysis (PCA) was applied to standardized numeric features to reduce dimensionality and summarize the data. The first five principal components explain approximately 63% of the total variance, with PC1 accounting for 28% and PC2 for 11%. Cumulative variance analysis indicates that around 10–11 components are required to capture 90% of the variance, suggesting moderate redundancy in the dataset. Overall, PCA effectively simplifies the data while retaining most of the information for subsequent analysis and modeling.

5.2. Clustering Analysis

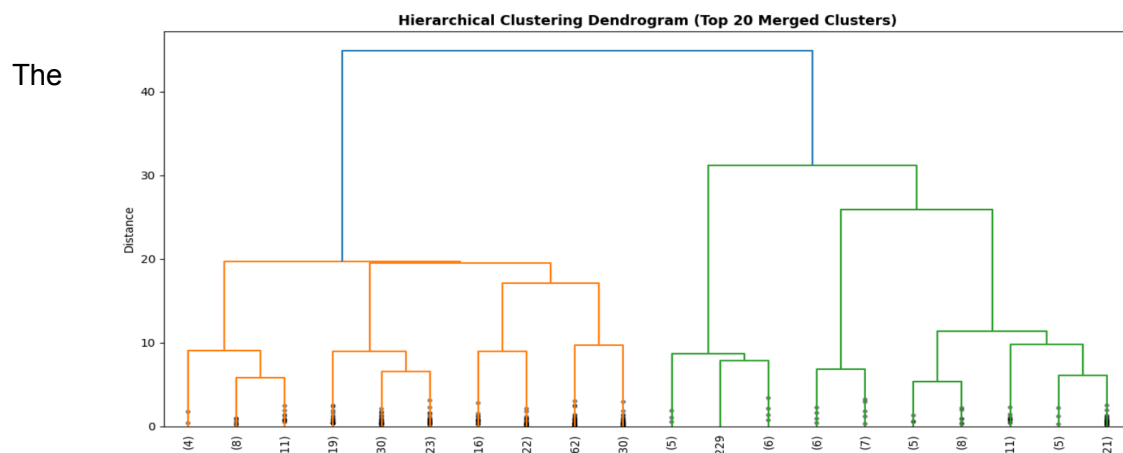
5.2.1 K-means Clustering

K-means clustering was applied to the PCA-reduced dataset using the first seven principal components, which explain approximately 68% of the total variance. The optimal number of clusters was identified using the elbow method and silhouette analysis, with results indicating that three clusters provide the best separation (silhouette score = 0.479). These clusters reveal meaningful patient subgroups with similar appointment and health characteristics for further analysis.



5.2.2 Hierarchical Clustering

Hierarchical clustering was applied to validate the cluster structure identified by K-means. Due to computational constraints, a random sample of 300 observations from the PCA-reduced dataset was used. Ward's linkage method and a dendrogram were employed to visualize cluster formation, followed by agglomerative clustering using the optimal number of clusters (k = 3).



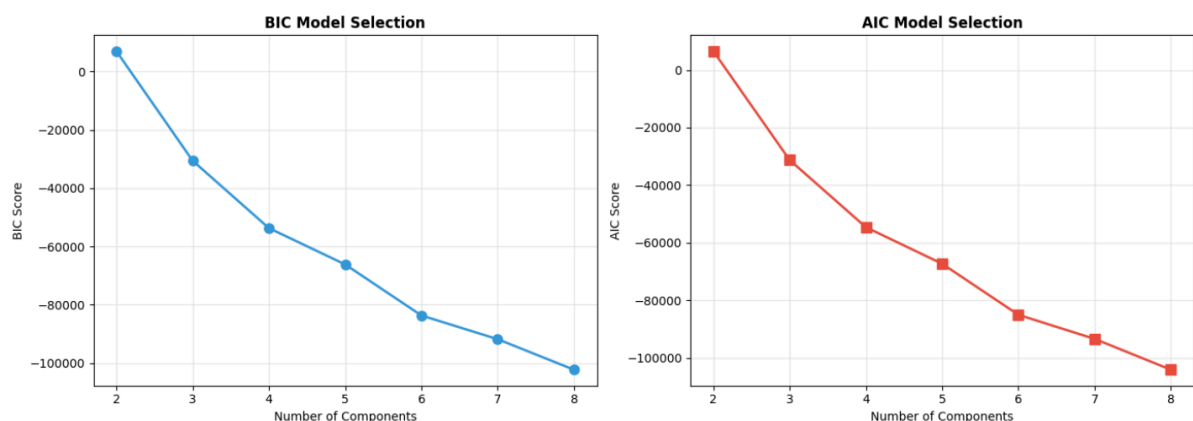
The hierarchical clustering results support the findings from K-means, confirming the presence of three main patient groups. The large dominant cluster likely represents typical appointment patterns, whereas the smaller clusters may correspond to specific patient profiles or

higher-risk groups. The dendrogram provides additional insight into how clusters are formed and highlights potential subgroups that could be further analyzed for targeted intervention strategies.

5.3. Gaussian Mixture Models (GMM)

Gaussian Mixture Models (GMM) were used to group patients based on probability, instead of strict distance like K-means. The model assumes that patient data come from several overlapping groups, where each group follows a Gaussian (normal) distribution.

The model was applied to the PCA-reduced data to reduce noise and correlation. For faster computation, a random sample of 5,000 patients was used. GMM models with 2 to 8 components were tested.



The GMM results suggest that patient behavior is more complex and overlapping than what is captured by K-means clustering. Instead of clear-cut groups, GMM identifies multiple patient profiles with shared characteristics. This indicates that no-show behavior may be influenced by several subtle patterns, rather than a small number of clearly separated groups. GMM therefore provides a more flexible view of patient segmentation in the no-show analysis.

5.3.1 Model Specification

This section explains which machine learning models are most suitable for predicting appointment no-shows, based on:

- The structure of the dataset
- The results from EDA, PCA, clustering, and statistical tests

Based on data characteristics and analysis results, tree-based models such as XGBoost and Random Forest are recommended for no-show prediction. These models handle mixed features, non-linear relationships, and class imbalance effectively. Performance should prioritize recall and AUC-ROC to better identify no-show patients.

Data Preparation for Modeling

Before training machine learning models, the dataset must be prepared properly:

1. Features and target
 - The target is **NoShow_binary** (1 = no-show, 0 = show).
 - Columns not needed for modeling, like IDs or original date columns, are dropped.
2. Identify feature types
 - Categorical features: 'Gender', 'Neighbourhood', 'AgeGroup', 'WaitingCategory', 'TimeOfDay'
 - Numerical features: age, waiting time, health conditions, etc.
3. Encode categorical features
 - Categorical columns are converted into numbers using LabelEncoder.
Example: 'Gender' → 0 (Female), 1 (Male)
 - This makes the data ready for models like XGBoost or Random Forest, which require numeric input.
4. Convert all columns to numeric
 - Ensures no text remains in the dataset.
5. Train-test split
 - The data is split into 80% train and 20% test.
 - Stratified split keeps the same proportion of no-show cases in both sets (~20%).

XGBoost Model

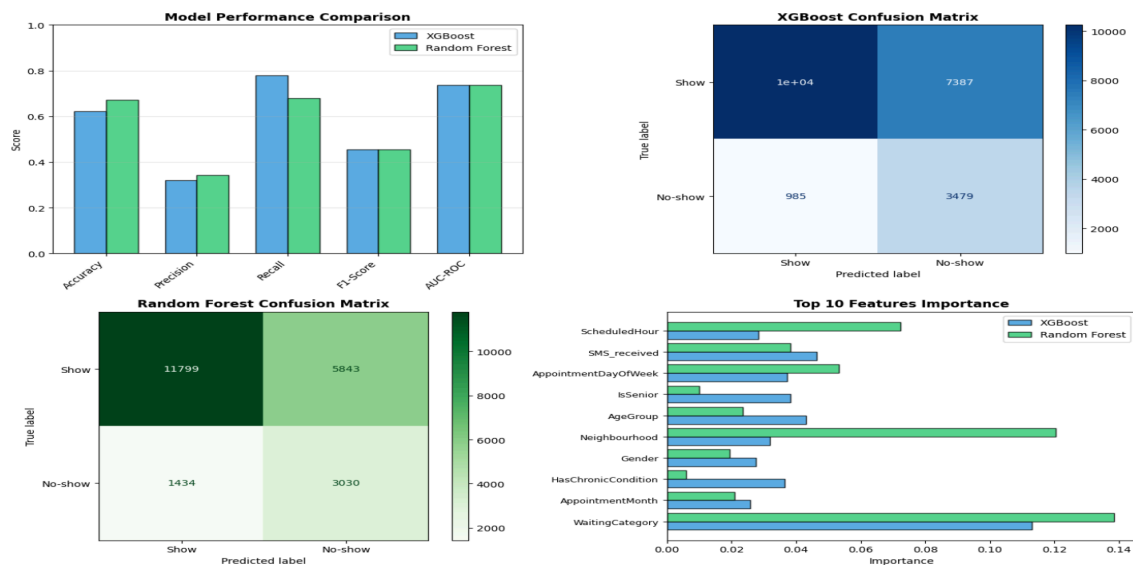
The XGBoost model was trained on the prepared dataset with class balancing to address the no-show class imbalance. The model achieved an accuracy of 62.5%, precision of 32.1%, recall of 76.7%, F1-score of 45.2%, and an AUC-ROC of 0.738, demonstrating strong performance in identifying no-show patients. Stratified 5-fold cross-validation confirmed stable performance with an average AUC-ROC of 0.732. Feature importance analysis identified `AwaitingDays` as the most influential predictor, followed by `WaitingCategory`, `SMS_received`, and `AgeGroup`, highlighting the importance of appointment timing and reminder-related factors.

Random Forest Model

The Random Forest model was trained on the prepared dataset using class weighting to address class imbalance. The model achieved an accuracy of 67.1%, precision of 34.0%, recall of 66.8%, F1-score of 45.1%, and an AUC-ROC of 0.735, indicating solid performance

in identifying no-show patients. Stratified 5-fold cross-validation confirmed stable results with an average AUC-ROC of 0.731. Feature importance analysis identified AwaitingDays as the most influential predictor, followed by WaitingCategory, Age, and Neighbourhood, highlighting the importance of appointment timing and patient demographics in no-show behavior.

Model Comparison



We compared XGBoost and Random Forest models for predicting patient no-shows using the same preprocessed dataset and evaluation metrics. XGBoost achieved higher recall (76.7%) and AUC-ROC (0.738), which are critical for identifying high-risk no-show patients, while Random Forest performed slightly better in accuracy (67.1%) and precision (34.0%). Both models showed similar F1-scores (~45%). Feature importance analysis indicated that AwaitingDays, WaitingCategory, and Age were the most influential predictors for both models. Overall, XGBoost demonstrated a better balance between detection capability and overall performance and is therefore selected as the preferred model for no-show prediction.

5.3.2 Component Interpretation

We used PCA to reduce the number of numeric features in our dataset while keeping most of the information. Each principal component is a combination of the original features that explains part of the differences in the data

- PC1 explains 28% of the variance, meaning it captures the largest patterns in the data. This component is mostly influenced by AwaitingDays and SMS_received, showing that the longer patients wait and whether they receive reminders is the most important factor in predicting no-shows.

- PC2 explains 11% of the variance and may capture patterns related to age and chronic conditions, meaning that patient age or health status is less important but still contributes to differences in attendance.
- PC3–PC5 explain smaller amounts of variance (7–8% each) and may represent other factors like appointment time, day of the week, or multiple conditions.
- Overall, PCA helps us simplify the dataset so that clustering and modeling focus on the most important patterns without losing much information.

After performing PCA, we reduced the dataset into key components that capture the most important patterns in patient behavior. Using these components, we grouped patients into clusters based on their similarities in features such as waiting time, age, health conditions, and SMS reminders. The optimal number of clusters was determined to be 3, using the silhouette score, which measures how well each patient fits within their assigned cluster compared to other clusters.

- Optimal number of clusters was 3, chosen using silhouette scores, which measure how well each patient fits into their cluster.
- Cluster 0: Mostly patients with short waiting times who showed up to appointments. This is the low-risk group.
- Cluster 1: Patients with long waiting times and higher chances of no-show. This is the high-risk group.
- Cluster 2: A small group with mixed features (maybe older patients or unusual appointment patterns).
- Hierarchical clustering confirms similar patterns: it shows how patients are merged into clusters based on similarity, and the dendrogram visualizes these relationships.
- The clusters help us identify which groups of patients are more likely to miss appointments, so interventions like reminders or rescheduling can be targeted.

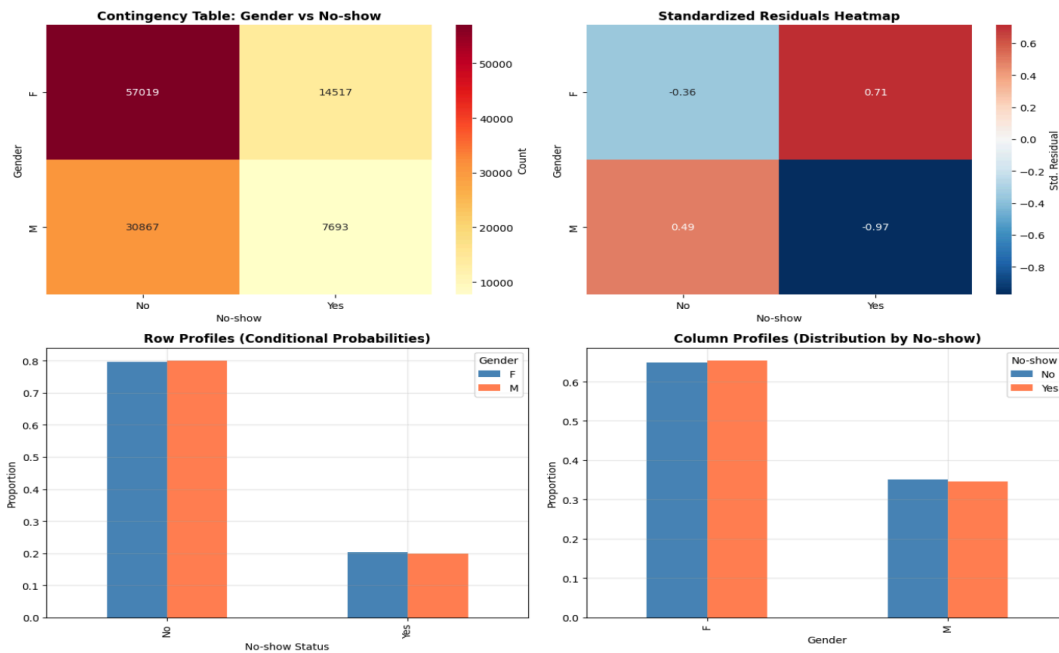
Gaussian Mixture Models (GMM) are similar to clustering, but instead of assigning each patient strictly to one group, GMM gives a probability of belonging to each group. This means we can understand patients who do not clearly fit into a single cluster, which is useful for real-world situations where behavior is mixed or uncertain.

- We found 8 components, each representing a group of patients with similar patterns.
- Some components represent high-risk no-show patients (e.g., long waiting days, missed SMS reminders).
- Other components represent low-risk patients (short waiting times, fewer health conditions).
- The probabilistic nature of GMM helps to understand uncertain cases, where patients might not clearly belong to just one cluster.

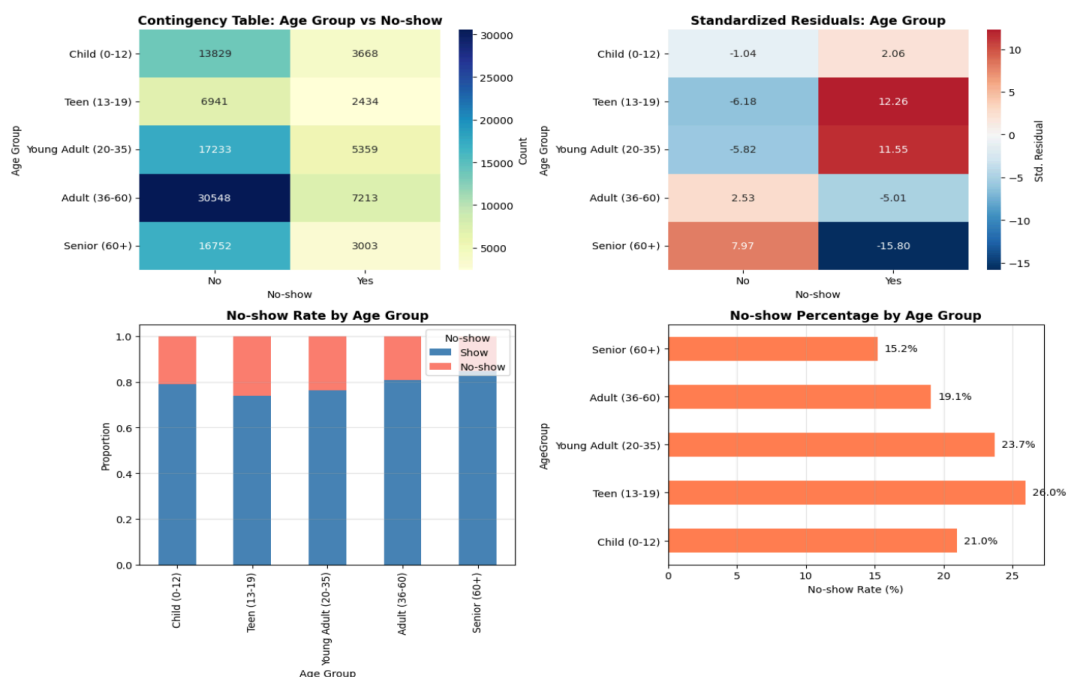
- These components give more detailed insights into patient behavior than simple clusters, helping hospitals plan better interventions.

5.4. Correspondence Analysis (CA)

5.4.1. Methodology Results



Correspondence analysis was used to examine the relationship between gender and appointment no-shows. Results showed a very weak association ($\chi^2 = 1.83$, $df = 1$), with standardized residuals close to zero for both genders, indicating observed counts were near expected values. No-show rates were nearly identical for females (20.3%) and males (19.9%), and column profiles showed no meaningful gender differences among no-show patients. Overall, gender has minimal influence on no-show behavior, suggesting that intervention strategies can be applied uniformly without gender-specific targeting.



Correspondence analysis was used to examine the relationship between patient demographics and no-show behavior, focusing on Gender and Age Group. Results showed that gender had little influence on no-show rates, with males and females displaying similar attendance patterns and a weak association. In contrast, age group exhibited a moderate relationship with no-show behavior: younger patients (especially ages 20–35) were more likely to miss appointments, while older adults and seniors showed higher attendance consistency. Row and column profile analyses supported these patterns, and standardized residuals identified age groups with higher- or lower-than-expected no-show rates. These findings suggest that intervention strategies should prioritize younger, high-risk age groups, while gender-neutral approaches are sufficient.

VI. Conclusion

This study examined hospital appointment data to understand and predict no-show behavior. PCA was used to reduce numeric features, while clustering methods (K-means, Hierarchical, and GMM) identified patient risk groups, with high-risk patients typically experiencing longer waiting times or missing reminders. Correspondence Analysis showed that age group is more strongly associated with no-shows than gender. For prediction, XGBoost and Random Forest models were applied, with XGBoost showing slightly better performance, particularly in recall for identifying high-risk patients. Key predictors included AwaitingDays, WaitingCategory, SMS reminders, and age-related variables, providing actionable insights to improve scheduling and reduce missed appointments.

Key Challenges:

- **Imbalanced Data:** Only ~20% of patients were no-shows, making prediction harder.
- **Mixed Data Types:** The dataset included numeric, categorical, and time features that needed careful preprocessing.
- **Complex Patient Behavior:** Some patients do not follow clear patterns, making prediction uncertain.
- **Feature Limitations:** Certain external factors (e.g., weather, transportation, personal reasons) were not in the dataset but may affect attendance.

Future Work:

- **Include More Features:** Integrate external data like weather, distance from hospital, or traffic conditions to improve prediction accuracy.
- **Advanced Models:** Explore deep learning models for complex pattern recognition.
- **Real-time Prediction:** Implement a system that predicts no-shows when patients book appointments to send reminders or alerts.

- Personalized Interventions: Design targeted strategies based on patient cluster or GMM component (e.g., extra reminders for high-risk groups).
- Broader Validation: Apply the model to other hospitals or departments to check its generalizability.

References

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Notebook name: data.ipynb

GitHub link:

https://github.com/Natt2509/EDA_project/blob/main/data.ipynb